

Bias Mitigation in ACSEmployment Prediction

CS340 Final Project

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Task and Evaluation Setup

Prediction Task

- Dataset: ACSEmployment, CA and TX, 2018
- Target: binary employment status
- Sensitive attribute: sex
- Held-out evaluation set: 129,383 individuals

Metric Scope

- 13 TFMA metrics
- 5 extended fairness metrics
- Focus: group-level reliability and error gaps

Goal: reduce gender-related fairness gaps without losing too much predictive utility.

Baseline Bias Pattern

Metric	Female	Male
AUC	0.8713	0.9244
Precision	0.7216	0.8267
Recall / TPR	0.8059	0.8853
FPR	0.2233	0.1787
FNR	0.1941	0.1147

Observed Gaps

- Predictive parity gap: 0.1051
- AUC gap: 0.0531
- Female group has higher FPR and FNR

Mitigation Target

Improve reliability and error-rate balance rather than only matching positive prediction rates.

Mitigation Method

1. Adversarial Debiasing

Gradient reversal discourages gender information in hidden representations.

2. Equalized Odds Loss

Directly penalizes FNR and FPR differences across groups.

3. Sample Re-weighting

Balances gender groups and boosts positive examples by 1.5x.

$$\mathcal{L}_{main} = \mathcal{L}_{BCE}(y, \hat{y}) + \lambda_{fair} (|FNR_f - FNR_m| + |FPR_f - FPR_m|)$$

$$\lambda_{adv} = 0.5, \quad \lambda_{fair} = 0.5, \quad \text{Adam optimizer, 25 epochs}$$

Extended Fairness Metrics Improved

Metric	Base Model	Fair Model	Direction
Disparate Impact Ratio	0.8886	0.9622	closer to 1
Average Odds Difference	0.0620	0.0330	lower
Predictive Parity Gap	0.1051	0.0971	lower
NPV Equality Gap	0.0335	0.0014	lower
AUC Gap	0.0531	0.0378	lower

All five extended fairness metrics move in the desired direction.

Performance Trade-off

Metric	Base	Fair	Change
AUC	0.9016	0.8784	-0.0232
Accuracy	0.8202	0.7922	-0.0280
Precision	0.7763	0.7066	-0.0697
Recall / TPR	0.8481	0.9265	+0.0784
FNR	0.1519	0.0735	-0.0784
FPR	0.2029	0.3193	+0.1164

Interpretation

- The model predicts more positives after mitigation.
- Fewer positive cases are missed.
- More false positives are introduced.

Main Trade-off

Fairness and recall improve, while accuracy, AUC, and precision decrease.

Overall Experimental Result

Improved Metrics

- 9 metrics improved in total
- 5 extended fairness metrics improved
- 4 TFMA metrics improved

TFMA Improvements

- Recall
- True positive rate
- False negative rate
- False omission rate

The improvement is targeted: fairness gaps and missed-positive errors improve, but predictive ranking and precision costs remain measurable.

Conclusion

- Adversarial debiasing, Equalized Odds loss, and sample re-weighting improve all five extended fairness metrics.
- The fair model reduces the AUC gap from 0.0531 to 0.0378 and predictive parity gap from 0.1051 to 0.0971.
- Recall improves from 0.8481 to 0.9265, reducing missed positive cases.
- Accuracy, AUC, precision, and FPR show that fairness gains come with real performance trade-offs.

Questions?